

KULDEEP SINGH

Automotive Embedded Systems & EV Integration Lead | Mobility Systems

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PROFESSIONAL SUMMARY

Automotive embedded systems and EV software leader targeting roles involving cross-domain automotive products, embedded software, hardware electronics integration, future vehicle architectures, and system integration. Brings 8 years of eVCU/BMS software, MATLAB/Simulink/Stateflow MBD, embedded C/C++, STM32/NXP/Infineon exposure, CAN diagnostics, HV battery systems, DCDC/OBC/PDU integration, charging controller logic, and vehicle-level validation. Strong fit for roles connecting hardware electronics, embedded software, software architecture, automotive product development, and global engineering delivery.

TARGET ROLE FIT

- System Integration Engineer - Automotive Applications
- Embedded Software Engineer - Automotive Products
- EV / Cross-Domain Vehicle Systems Engineer
- BMS / ECU Software Integration Lead

CORE STRENGTHS

- System integration fit: developed cross-domain EV systems linking hardware electronics, embedded software, CAN communication, power electronics, battery systems, vehicle controls, and validation.
- Embedded systems fit: developed STM32-based instrument cluster, dual-CAN telematics, ultra-capacitor monitoring/control, LV/HV PDU, and eVCU/BMS application software.
- Automotive product fit: delivered SOR, supplier coordination, validation evidence, quality tools, and ICAT/ARAI homologation readiness for commercial EV and hybrid programmes.

PROFESSIONAL EXPERIENCE

Technical Manager - Product Development

IX Energy Pvt. Ltd., Noida, India | Jul 2018 - Present

- Integrated hardware electronics and embedded software across HV battery, BMS/eVCU, DCDC, OBC, PDU, motor controller, telematics, sensors, and vehicle controls.
- Developed MATLAB/Simulink/Stateflow eVCU and BMS application software for functional requirements, safety requirements, diagnostics, fault handling, and safe vehicle operating states.
- Built STM32F4/F1 instrument cluster, Quectel dual-CAN telematics, LV/HV PDU, and ultra-capacitor cell monitoring/control system for in-house EV component development.
- Defined component/system specifications and SOR documents for supplier sourcing and integration across battery, motor, gearbox, DCDC, EPS, BCS, TCS, HVAC, and controller systems.
- Used CAN tools, diagnostics, DFMEA, RCA, Six Sigma, ISO 26262 awareness, and IATF 16949 process thinking to support robust automotive product development.

SELECTED PROGRAMMES & PROOF

- LCV/LPT EV conversion: ARAI-homologated commercial EV with eVCU/BMS logic, vehicle control, charging interfaces, DCDC/OBC/PDU integration, diagnostics, and HV battery coordination.
- Embedded component suite: delivered LV/HV PDU, STM32 instrument cluster, dual-CAN telematics, ultra-capacitor monitoring/control, and battery-related software interfaces.
- HV battery systems: packaged and integrated 5 kWh to 300 kWh packs across LFP, NMC, LTO, and ultra-capacitor technologies with BTMS/HVAC validation.

TECHNICAL TOOLKIT & CERTIFICATIONS

Embedded C/C++, STM32, Microchip PIC, NXP, Infineon, MATLAB, Simulink, Stateflow, BMS, eVCU, CAN, Vector CANoe, Peak CAN, Bus Master, CSS Electronics, DBC, diagnostics, software architecture, system integration, ISO 26262, IATF 16949.

- Six Sigma Black Belt - Certified
- ISO 26262 Functional Safety for Road Vehicles - TUV SUD Certified
- IATF 16949:2016 Automotive Quality Management - TUV SUD Certified
- AWS IoT & SIMULINK - Amazon Web Services / MathWorks Certified

EDUCATION

B.Tech - Mechanical Engineering, VIT University, Vellore, Tamil Nadu | 2014 - 2018 | First Class, 75%